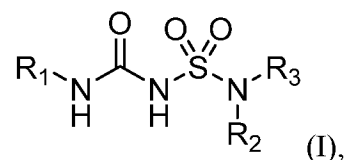


CLAIMS:

1. A compound of Formula (I):



or a solvate, or pharmaceutically acceptable salt thereof, wherein:

R₁ is C₃-C₁₆ cycloalkyl, or C₅-C₁₀ aryl, wherein the C₃-C₈ monocyclic cycloalkyl, polycyclic cycloalkyl, or C₅-C₆ aryl is optionally substituted by one or more **R_{1S}**; wherein each **R_{1S}** is independently C₁-C₆ alkyl, C₁-C₆ haloalkyl, C₁-C₆ alkoxy, C₁-C₆ haloalkoxy, or halo;

R₂ is -(C**X₂****X₂**)_n-**R_{2S}**, wherein n is 0, 1, or 2, and each **X₂** is independently H, C₁-C₆ alkyl, C₂-C₆ alkenyl, or C₂-C₆ alkynyl, wherein the C₁-C₆ alkyl, C₂-C₆ alkenyl, or C₂-C₆ alkynyl is optionally substituted with one or more halo, -CN, -OH, -O(C₁-C₆ alkyl), -NH₂, -NH(C₁-C₆ alkyl), -N(C₁-C₆ alkyl)₂, or oxo;

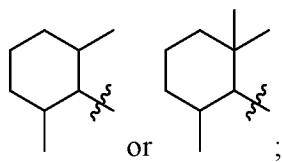
R_{2S} is 4- to 8-membered heterocycloalkyl optionally substituted with one or more C₁-C₆ alkyl, C₂-C₆ alkenyl, C₂-C₆ alkynyl, C₁-C₆ haloalkyl, halo, -CN, -OH, -O(C₁-C₆ alkyl), -NH₂, -NH(C₁-C₆ alkyl), -N(C₁-C₆ alkyl)₂, or oxo; and

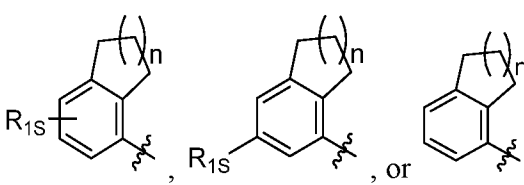
R₃ is 7- to 12-membered heterocycloalkyl or 5- or 6-membered heteroaryl optionally substituted with one or more **R_{3S}**, wherein each **R_{3S}** is independently C₁-C₆ alkyl, C₁-C₆ haloalkyl, C₃-C₈ cycloalkyl, halo, or C₃-C₈ heterocycloalkyl wherein the C₁-C₆ alkyl, C₁-C₆ haloalkyl, C₃-C₈ cycloalkyl, or C₃-C₈ heterocycloalkyl is optionally substituted with -O(C₁-C₆ alkyl), -N(C₁-C₆ alkyl)₂, halo, or -CN.

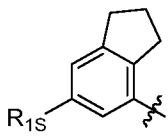
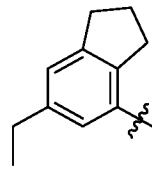
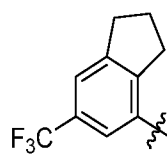
2. The compound of claim 1, wherein **R₁** is C₃-C₇ monocyclic cycloalkyl, C₉-C₁₀ bicyclic cycloalkyl, C₁₂-C₁₆ tricyclic cycloalkyl, or C₅-C₁₀ aryl, wherein the C₃-C₇ monocyclic cycloalkyl, C₉-C₁₀ bicyclic cycloalkyl, C₁₂-C₁₆ tricyclic cycloalkyl, or C₅-C₁₀ aryl, is optionally substituted by one or more **R_{1S}**;

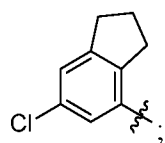
optionally wherein:

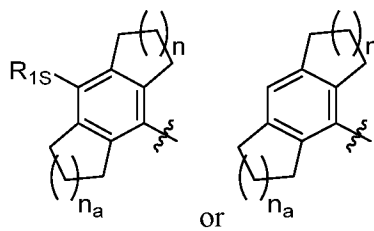
(a) the C₃-C₇ monocyclic cycloalkyl is cyclopentyl, cyclohexyl, or cycloheptyl, wherein the cyclopentyl, cyclohexyl, or cycloheptyl is optionally substituted by one or more **R**_{1s}; further

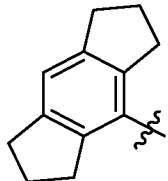
optionally wherein the C₃-C₇ monocyclic cycloalkyl is  or ;

(b) the C₉-C₁₀ bicyclic cycloalkyl is , wherein n

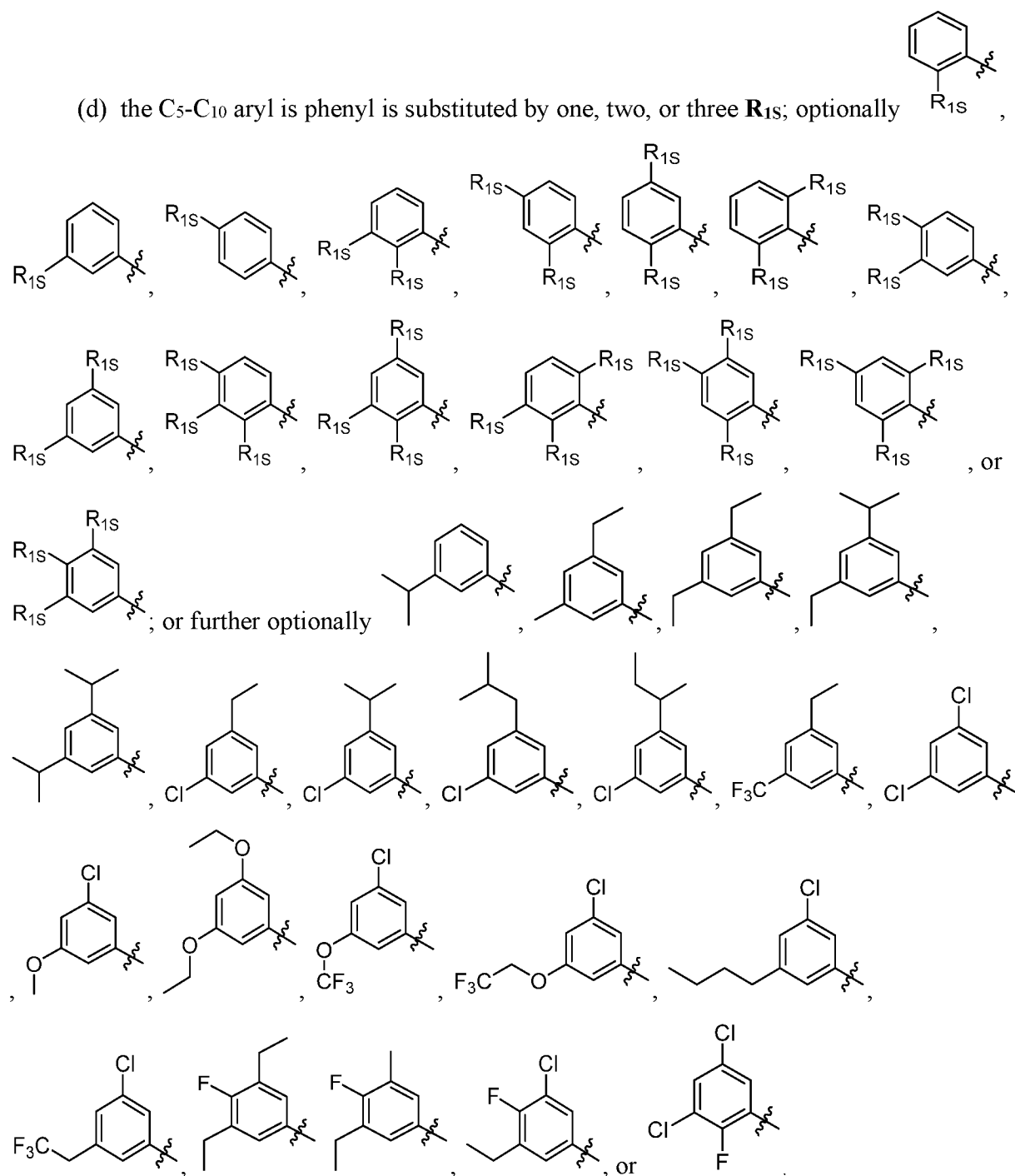
is 0, 1, or 2; optionally ; further optionally , , or



(c) the C₁₂-C₁₆ tricyclic cycloalkyl is , wherein n and n_a

each independently are 0, 1, 2, or 3; further optionally ; and/or

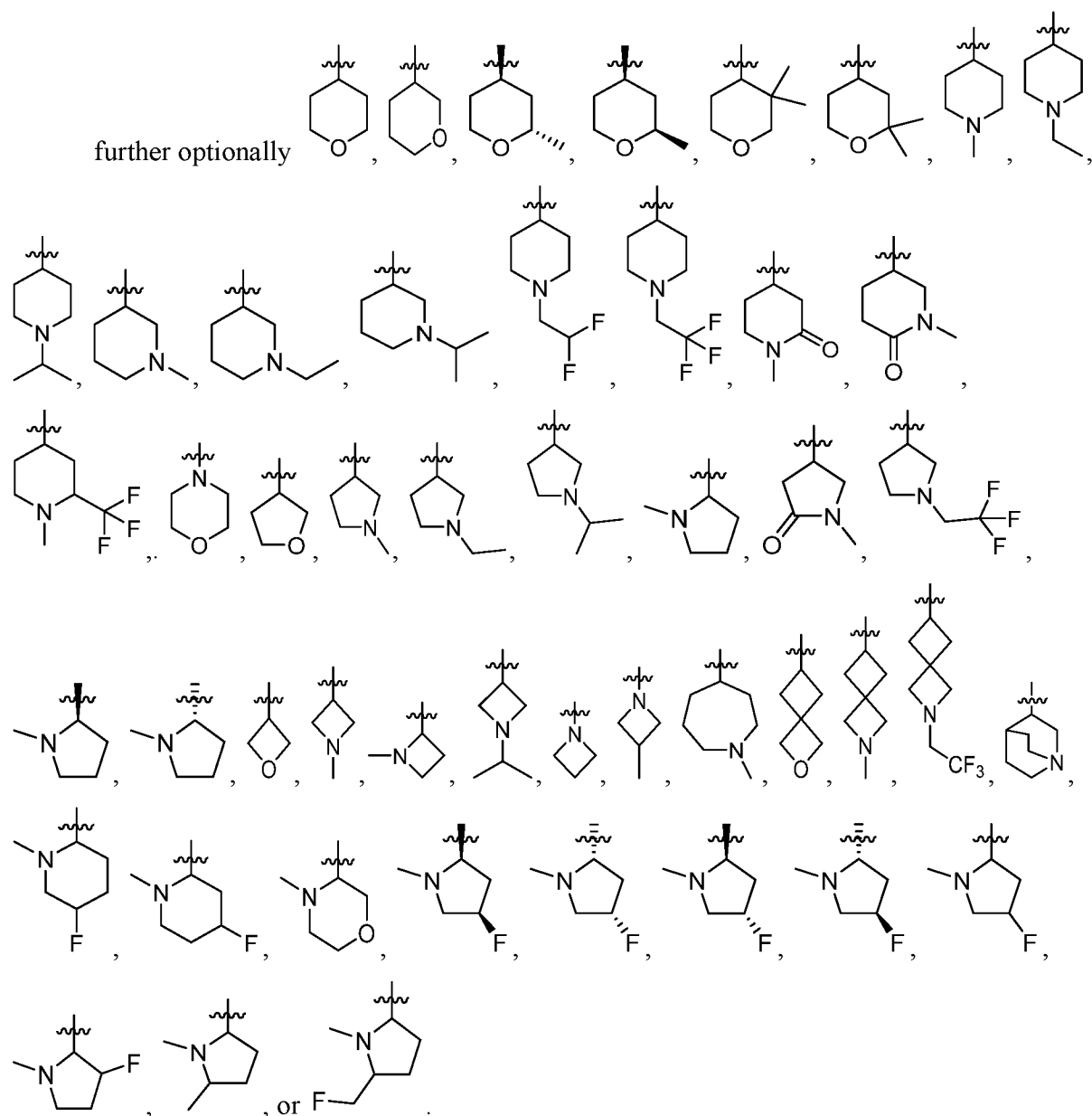
(d) the C₅-C₁₀ aryl is phenyl is substituted by one, two, or three **R**_{1s}; optionally $\overset{\text{R}_{1s}}{\text{I}}$,



3. The compound of claim 1 or 2, wherein **R**₂ is -**R**_{2S}, -(CX₂X₂)-**R**_{2S}, or -(CX₂X₂)₂-**R**_{2S}.
4. The compound of any one of claims 1 to 3, wherein **R**_{2S} is 4- to 8-membered heterocycloalkyl optionally substituted with one or more C₁-C₆ alkyl, C₂-C₆ alkenyl, C₂-C₆

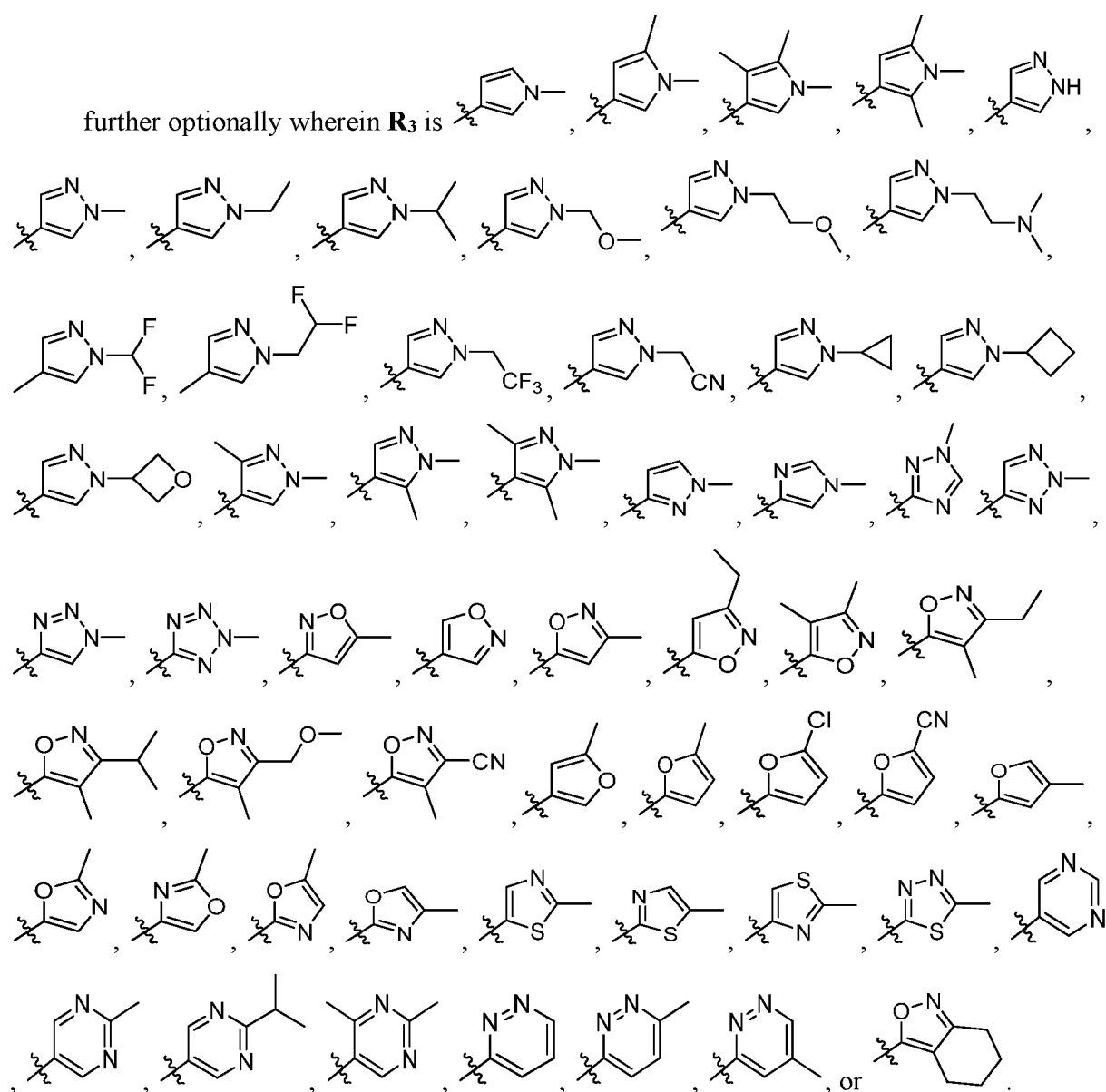
alkynyl, C₁-C₆ haloalkyl, halo, -CN, -OH, -O(C₁-C₆ alkyl), -NH₂, -NH(C₁-C₆ alkyl), -N(C₁-C₆ alkyl)₂, or oxo;

optionally oxetanyl, azetidiny, tetrahydrofuranyl, pyrrolidinyl, tetrahydropyranyl, piperidinyl, morpholinyl, azepanyl, 2-azaspiro[3.3]heptane, 2-oxaspiro[3.3]heptane, or bridged quinuclidinyl, wherein the oxetanyl, azetidiny, tetrahydrofuranyl, pyrrolidinyl, tetrahydropyranyl, piperidinyl, morpholinyl, azepanyl, 2-azaspiro[3.3]heptane, 2-oxaspiro[3.3]heptane, or bridged quinuclidinyl is optionally substituted with one or more C₁-C₆ alkyl, C₁-C₆ haloalkyl, or oxo; or

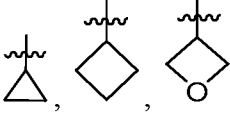


5. The compound of any one of claims 1 to 4, wherein **R₃** is 7- to 12-membered heterocycloalkyl optionally substituted with one or more **R_{3s}**;

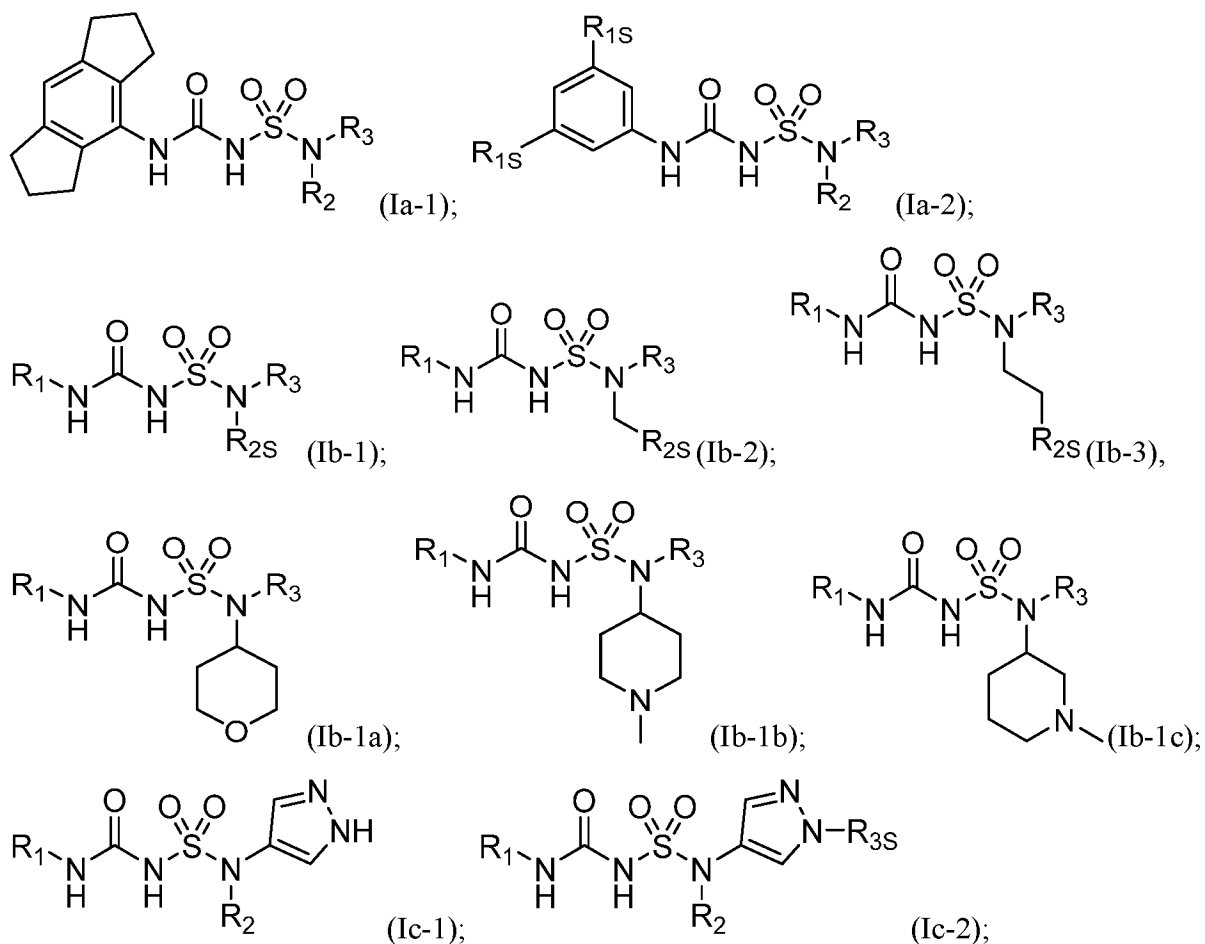
optionally wherein **R₃** is pyrrolyl, pyrazolyl, imidazolyl, triazolyl, tetrazolyl, isoxazolyl, furanyl, oxazolyl, isothiazolyl, thiadiazolyl, pyridinyl, pyrimidinyl, pyridazinyl, or 4,5,6,7-tetrahydrobenzo[c]isoxazole, wherein the pyrrolyl, pyrazolyl, imidazolyl, triazolyl, tetrazolyl, isoxazolyl, furanyl, oxazolyl, isothiazolyl, thiadiazolyl, pyridinyl, pyrimidinyl, pyridazinyl, or 4,5,6,7-tetrahydrobenzo[c]isoxazole is optionally substituted with one or more **R_{3s}**; or

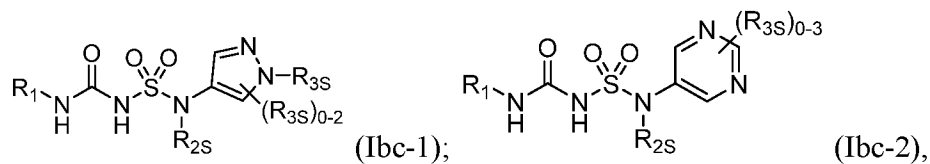
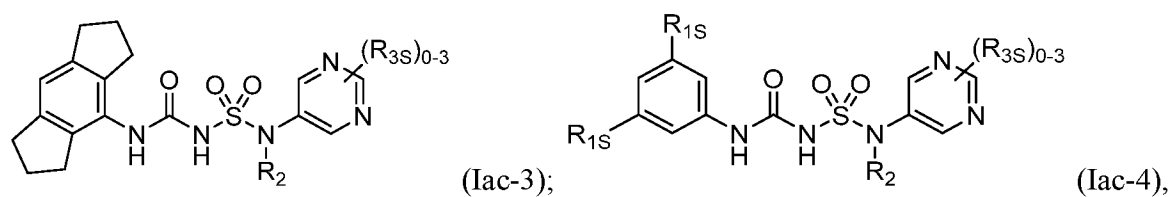
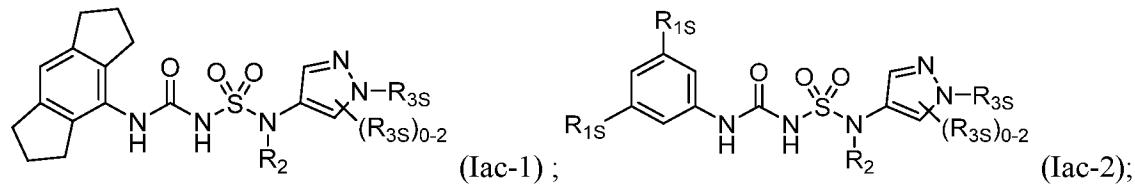
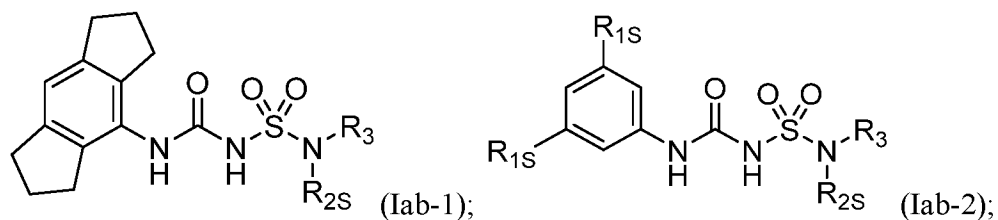
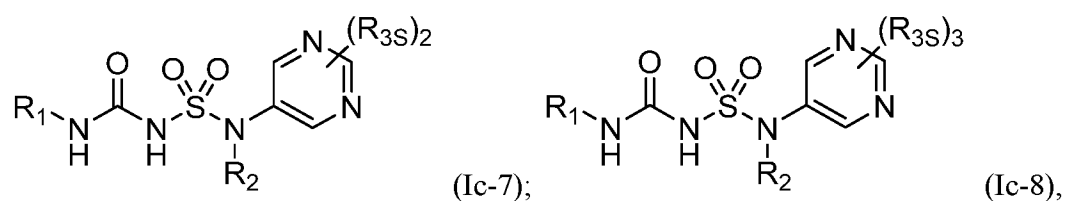
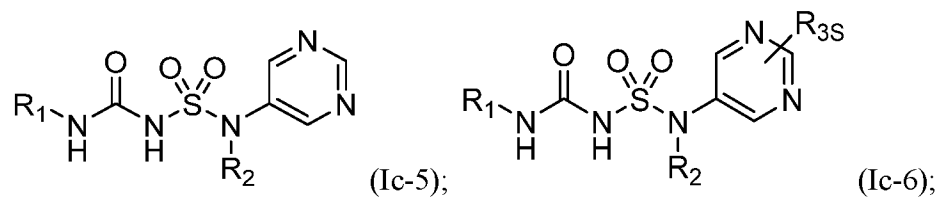
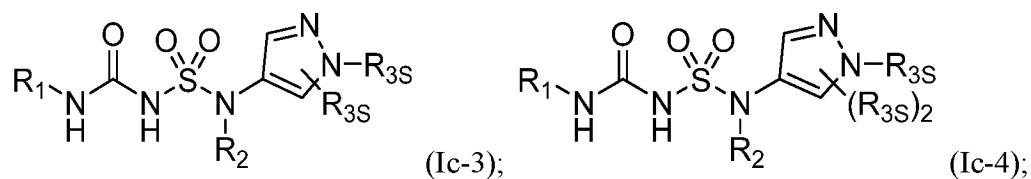


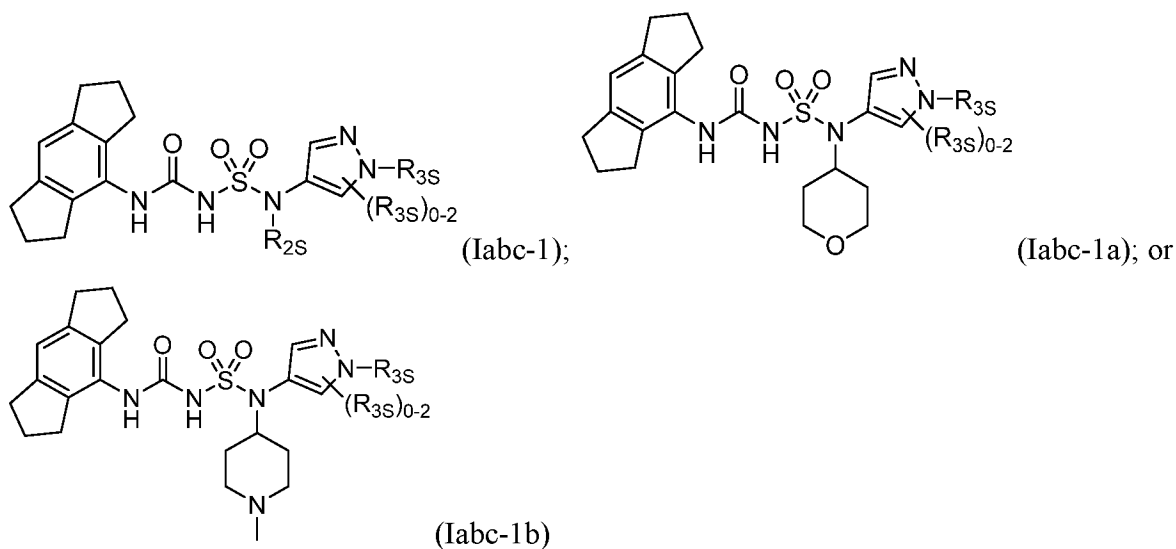
6. The compound of any one of claims 1 to 5, wherein at least one **R**_{3s} is C₁-C₆ alkyl, C₁-C₆ haloalkyl, C₃-C₈ cycloalkyl, halo, or C₃-C₈ heterocycloalkyl wherein the C₁-C₆ alkyl, C₁-C₆ haloalkyl, C₃-C₈ cycloalkyl, or C₃-C₈ heterocycloalkyl is optionally substituted with -O(C₁-C₆ alkyl), -N(C₁-C₆ alkyl)₂, halo, or -CN;

optionally wherein at least one **R**_{3s} is , methyl, ethyl, isopropyl, -CH₂OCH₃, -CH₂CF₃, -CH₂CH₂OCH₃, -CH₂CN, CH₂CF₂, or -CH₂CH₂N(CH₃)₂.

7. The compound of any one of the preceding claims, wherein the compound is of Formula (Ia-1), (Ia-2), (Ib-1), (Ib-2), (Ib-3), (Ib-1a), (Ib-1b), (Ib-1c), (Ic-1), (Ic-2), (Ic-3), (Ic-4), (Ic-5), (Ic-6), (Ic-7), (Ic-8), (Iab-1), (Iab-2), (Iac-1), (Iac-2), (Iac-3), (Iac-4), (Ibc-1), (Ibc-2), (Iabc-1), (Iabc-1a), or (Iabc-1b):

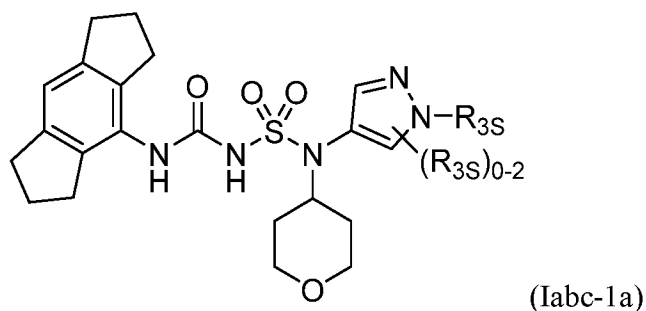






or a solvate, or pharmaceutically acceptable salt thereof, wherein **R₁**, **R_{2S}**, **R₂**, and **R₃** are as described herein.

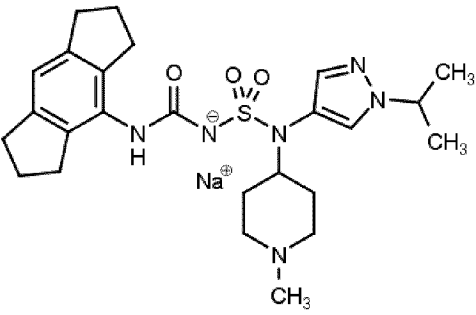
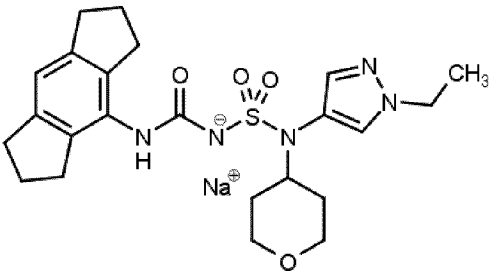
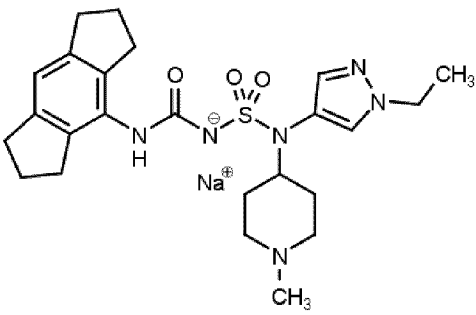
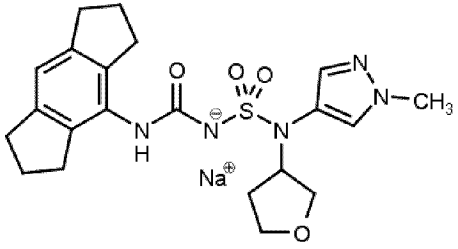
8. The compound of any one of claims 1 to 7, wherein the compound is of Formula (Iabc-1a):



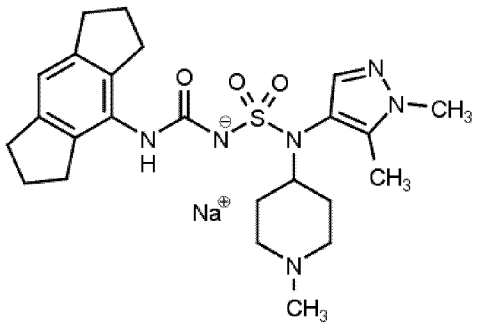
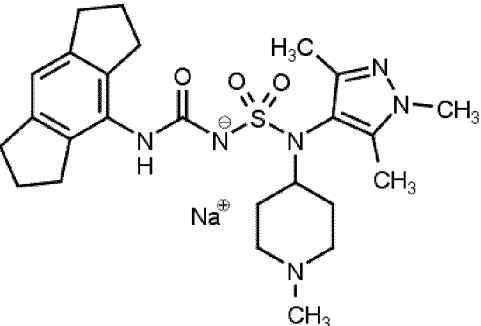
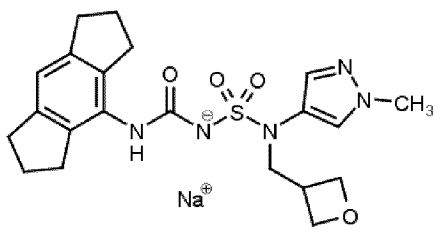
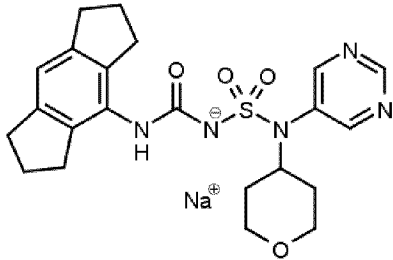
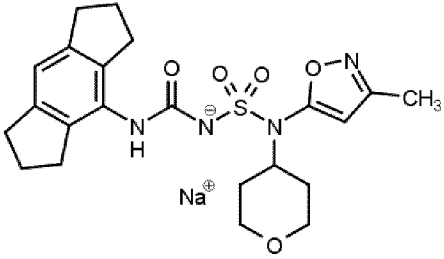
or a solvate, or pharmaceutically acceptable salt thereof, wherein **R_{3S}** is as described herein.

9. The compound of any one of the claims 1 to 8, being selected from Compound Nos. 1-166:

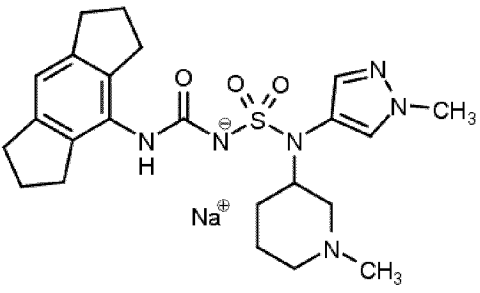
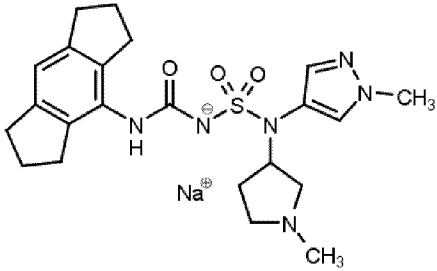
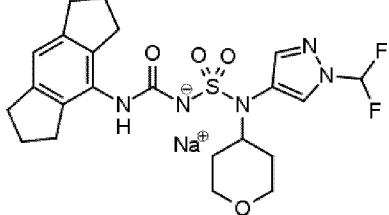
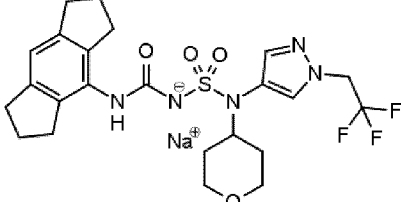
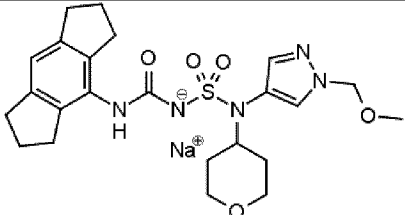
Compound No.	Structure
1	 <chem>CN1C=CC=C1N(C2CCOCC2)S(=O)(=O)N(C(=O)N3C4C=CC5C4C=CC3C5)C6=CC=CC=C6</chem>
2	 <chem>CN1C=CC=C1N(C2CCN(C)CC2)S(=O)(=O)N(C(=O)N3C4C=CC5C4C=CC3C5)C6=CC=CC=C6</chem>
3	 <chem>CN1C=CC=C1N(C2CCN(C)CC2)S(=O)(=O)N(C(=O)N3C4C=CC5C4C=CC3C5)C6=CC=CC=C6</chem>
4	 <chem>CC(C)N1C=CC=C1N(C2CCOCC2)S(=O)(=O)N(C(=O)N3C4C=CC5C4C=CC3C5)C6=CC=CC=C6</chem>

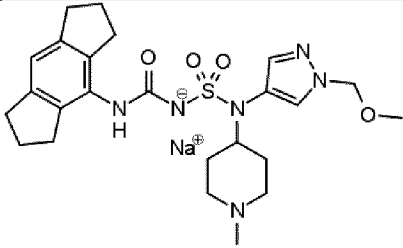
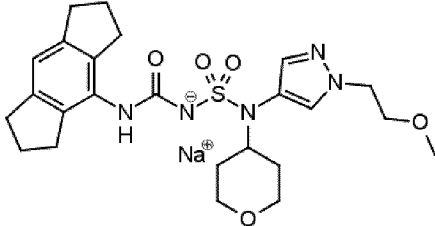
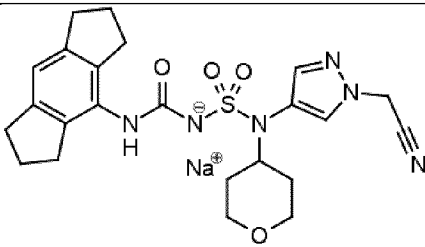
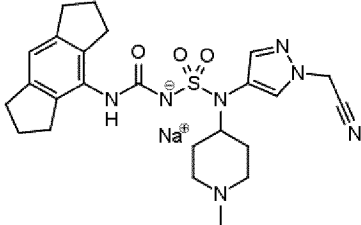
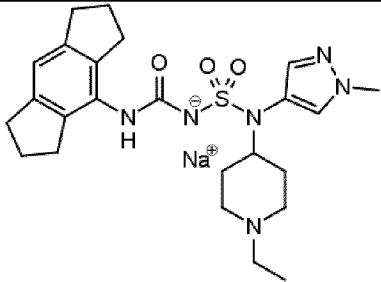
Compound No.	Structure
5	
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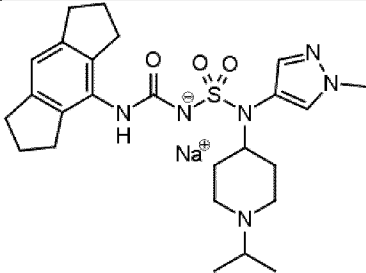
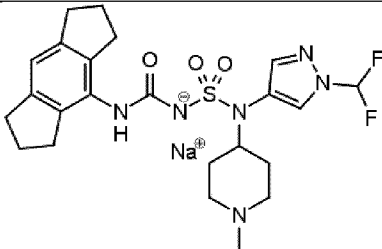
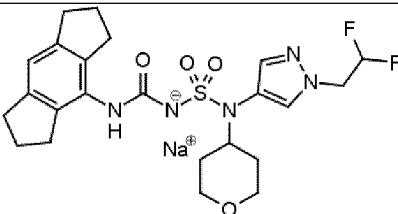
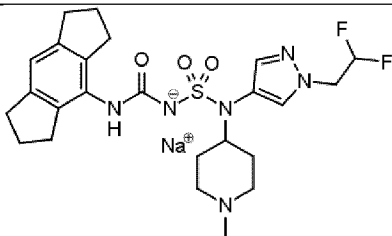
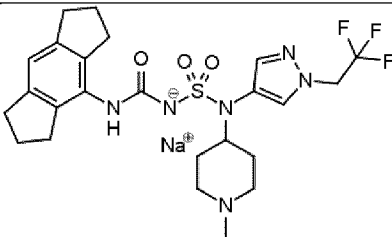
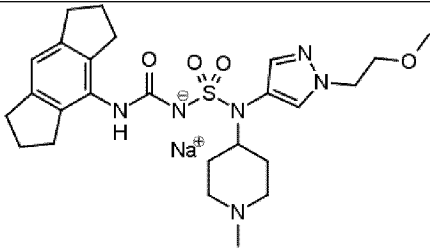
Compound No.	Structure
9	 <chem>CN1C=CN(C1)C2(COCCO2)N(S(=O)(=O)[N-]C(=O)N3C4C=CC5C4C=CC5C3)[S+](=O)(=O)[O-].[Na+]</chem>
10	 <chem>CN1C=CN(C1C)C2(COCCO2)N(S(=O)(=O)[N-]C(=O)N3C4C=CC5C4C=CC5C3)[S+](=O)(=O)[O-].[Na+]</chem>
11	 <chem>CN1C=CN(C1C)C2(COCCO2)N(S(=O)(=O)[N-]C(=O)N3C4C=CC5C4C=CC5C3)[S+](=O)(=O)[O-].[Na+]</chem>
12	 <chem>CC1=C(C2=CC=CC2=N1)N(S(=O)(=O)[N-]C(=O)N3C4C=CC5C4C=CC5C3)[S+](=O)(=O)[O-].[Na+]</chem>
13	 <chem>C1=CC2=CC=CC2=N1C3(COCCO3)N(S(=O)(=O)[N-]C(=O)N4C5C=CC6C5C=CC6C4)[S+](=O)(=O)[O-].[Na+]</chem>

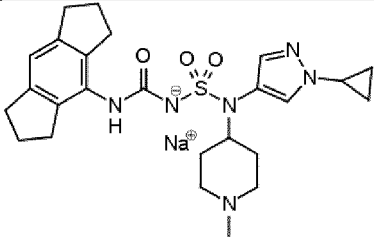
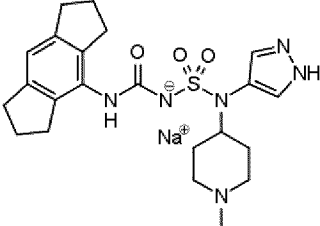
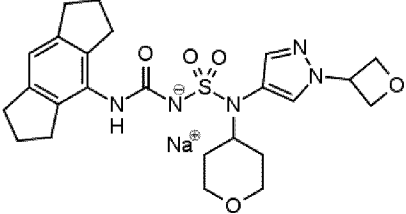
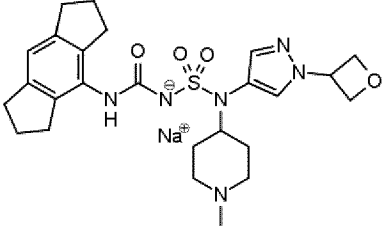
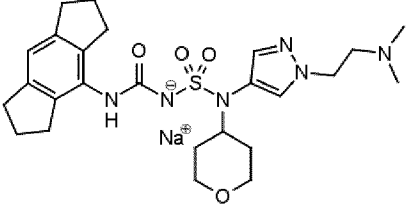
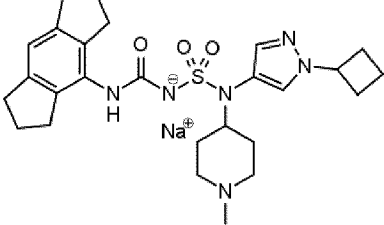
Compound No.	Structure
14	
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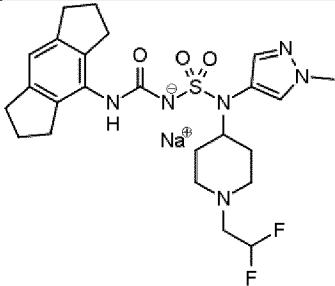
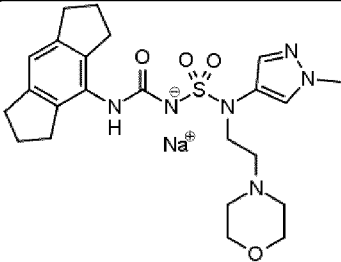
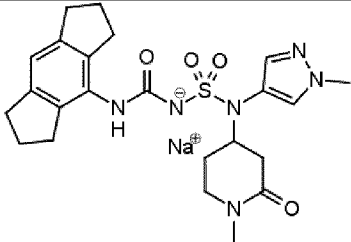
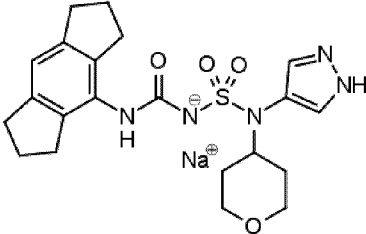
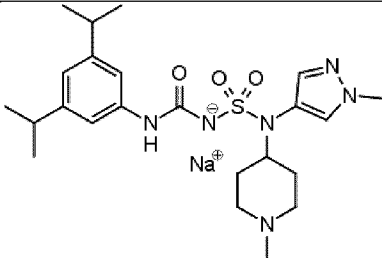
Compound No.	Structure
19	 <chem>CN1C=NC=C1N(S(=O)(=O)NC(=O)N2C3C=CC4C(C3)CC4)C5C=CCO5.[Na+]</chem>
20	 <chem>CN1C=CN=C1N(S(=O)(=O)NC(=O)N2C3C=CC4C(C3)CC4)C5CCN(C)CC5.[Na+]</chem>
21	 <chem>CN1C=CN=C1N(S(=O)(=O)NC(=O)N2C3C=CC4C(C3)CC4)C5C=CCO5.[Na+]</chem>
22	 <chem>CN1C=CN=C1N(S(=O)(=O)NC(=O)N2C3C=CC4C(C3)CC4)C5C=CCO5.[Na+]</chem>

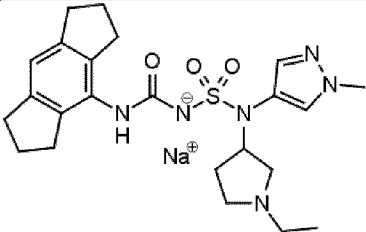
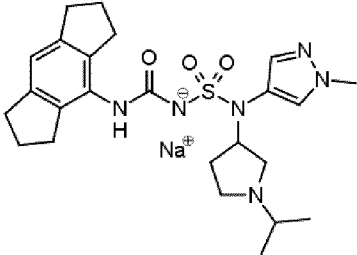
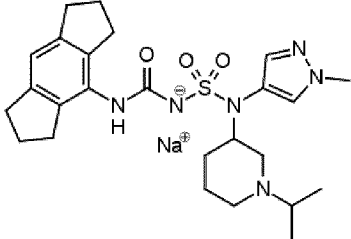
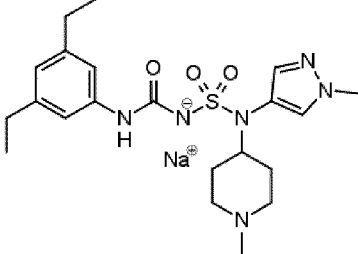
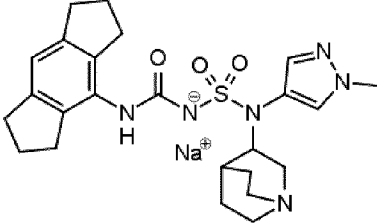
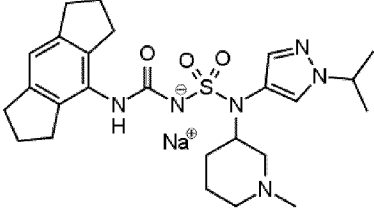
Compound No.	Structure
23	
24	
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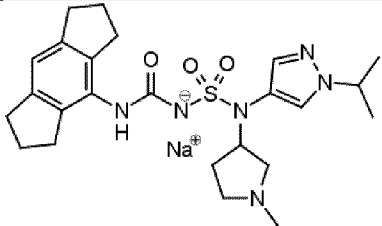
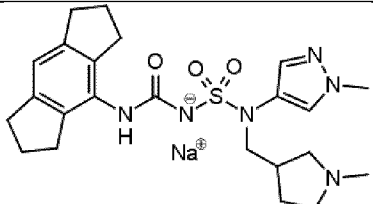
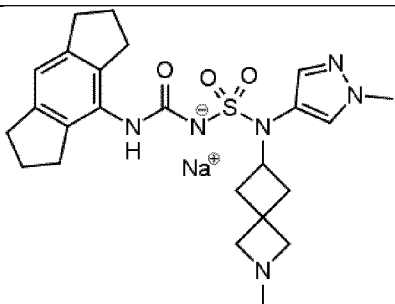
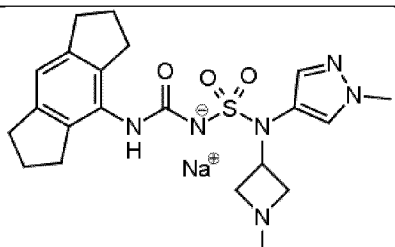
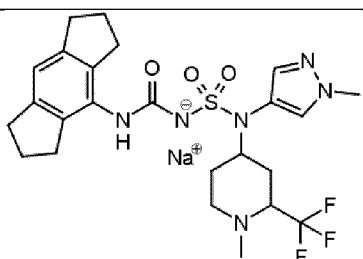
Compound No.	Structure
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Compound No.	Structure
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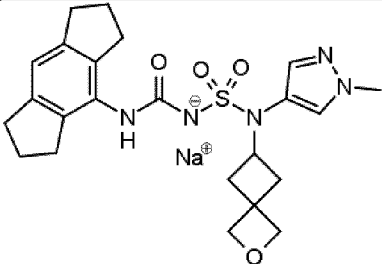
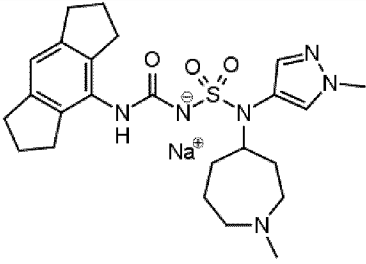
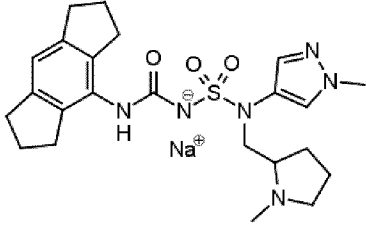
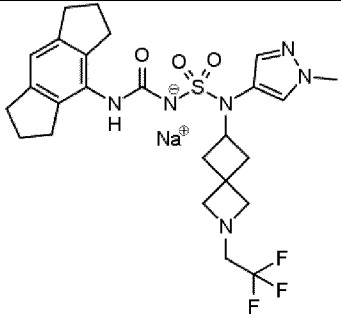
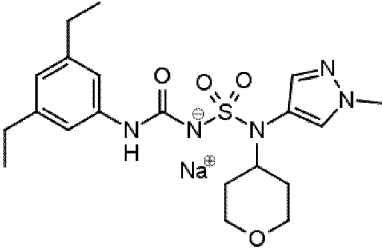
Compound No.	Structure
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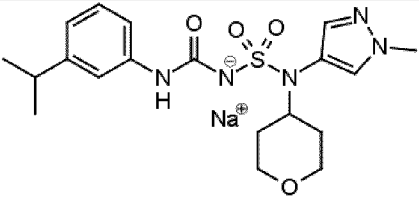
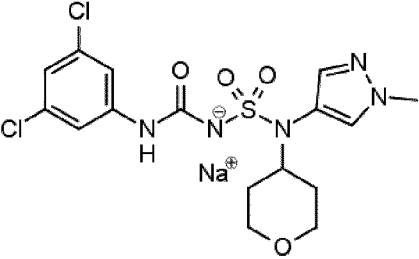
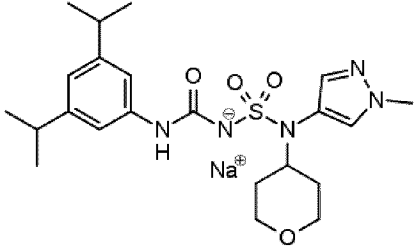
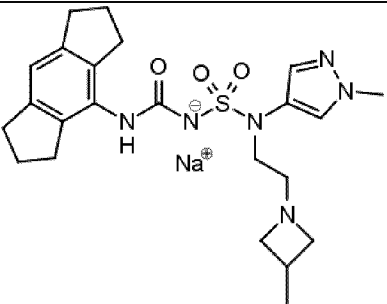
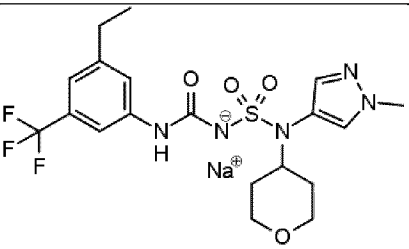
Compound No.	Structure
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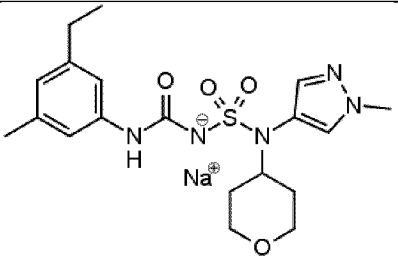
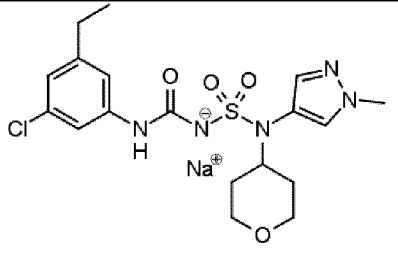
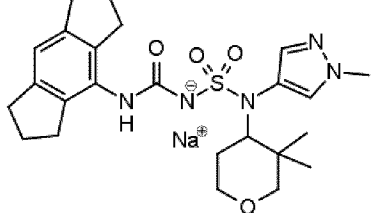
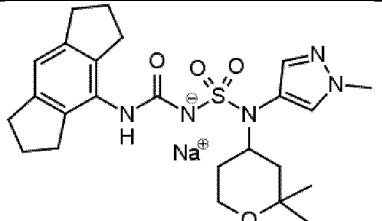
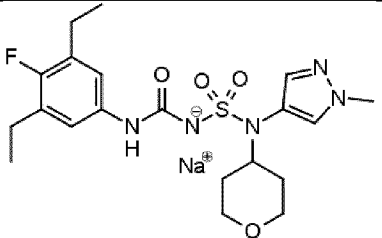
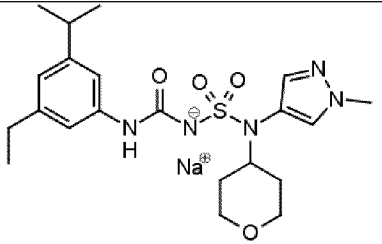
Compound No.	Structure
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Compound No.	Structure
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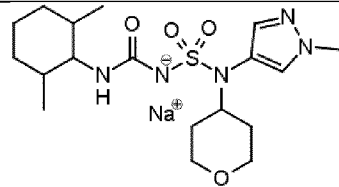
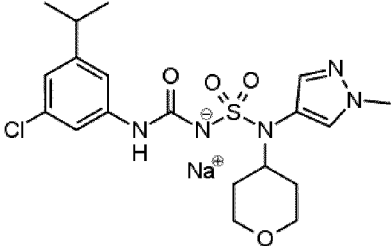
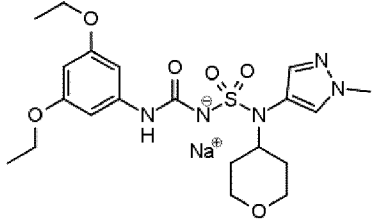
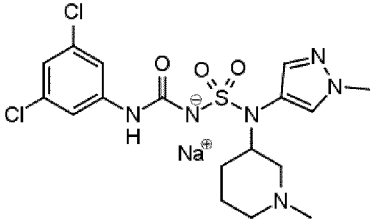
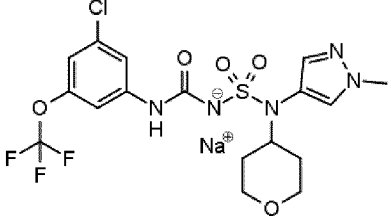
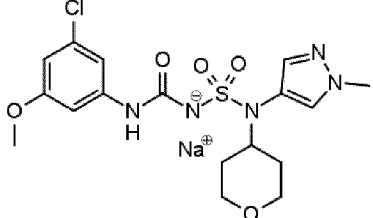
Compound No.	Structure
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62	 <chem>CCN1CCCCC1N(S(=O)(=O)N2C(=O)Nc3ccc4C(C2)CCc4c3)c5cc(C)nn5[Na+]</chem>
63	 <chem>C1CCN1CCN(S(=O)(=O)N2C(=O)Nc3ccc4C(C2)CCc4c3)c5cc(C)nn5[Na+]</chem>
64	 <chem>CC(C)N1CCN(C1)S(=O)(=O)N2C(=O)Nc3ccc4C(C2)CCc4c3)c5cc(C)nn5[Na+]</chem>
65	 <chem>C1CCN(CC1)CC(F)(F)FN(S(=O)(=O)N2C(=O)Nc3ccc4C(C2)CCc4c3)c5cc(C)nn5[Na+]</chem>

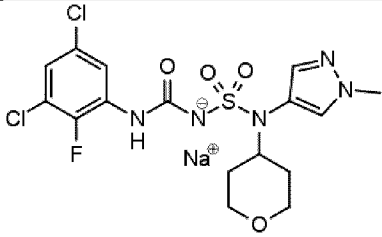
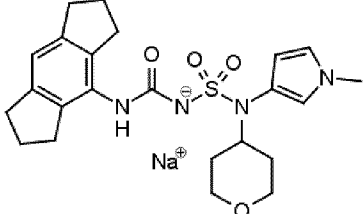
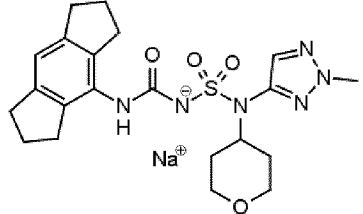
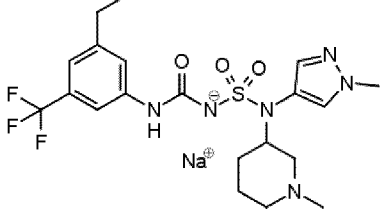
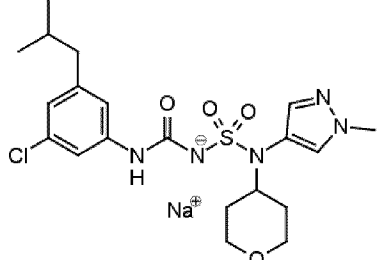
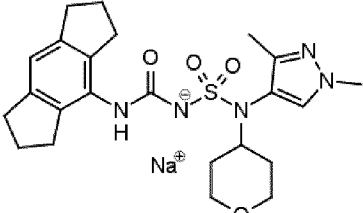
Compound No.	Structure
66	
67	
68	
69	
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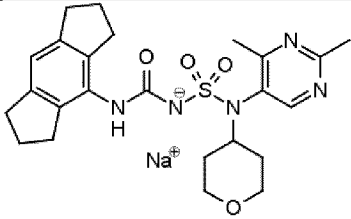
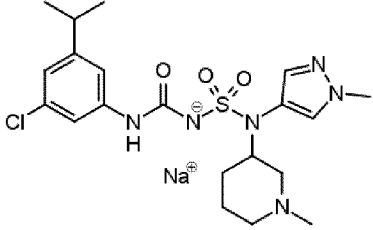
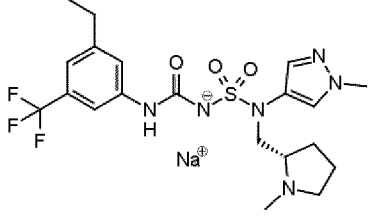
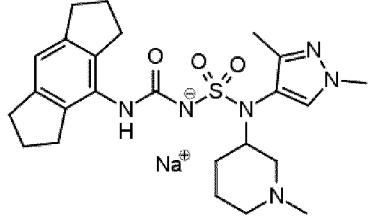
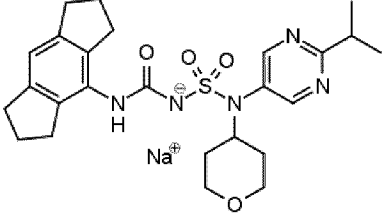
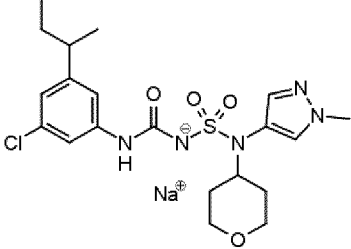
Compound No.	Structure
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72	
73	
74	
75	

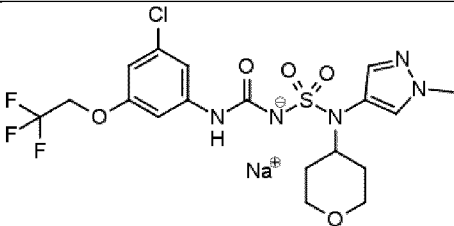
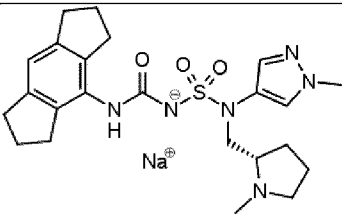
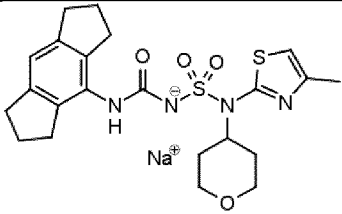
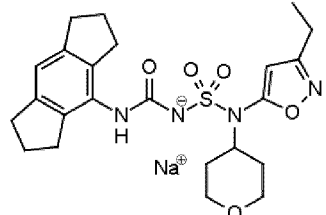
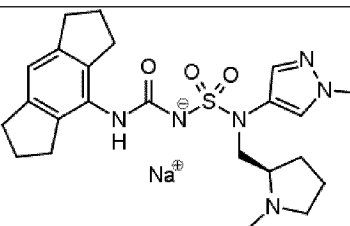
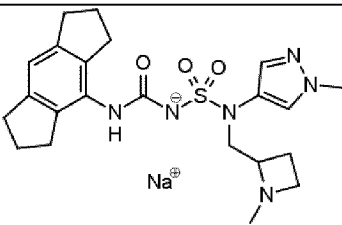
Compound No.	Structure
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77	
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81	

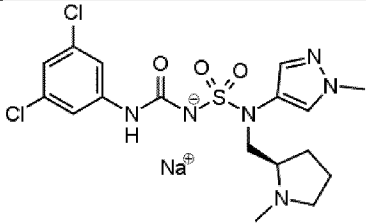
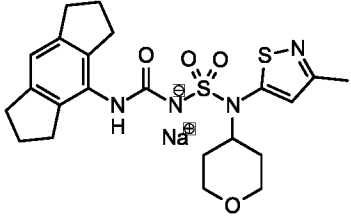
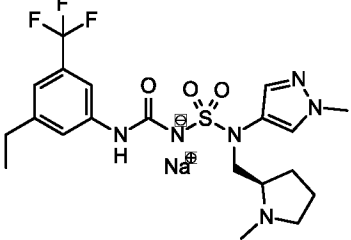
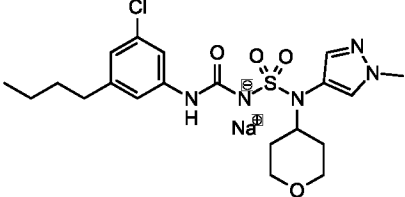
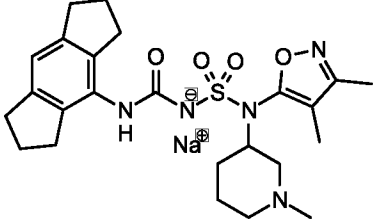
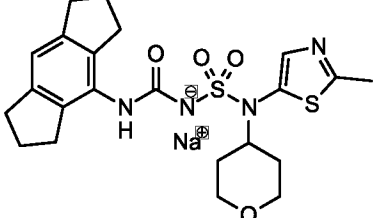
Compound No.	Structure
82	
83	
84	
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86	
87	

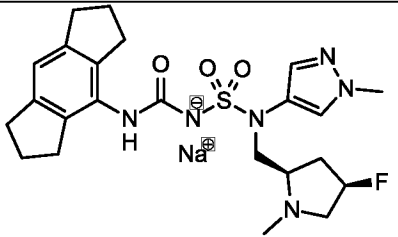
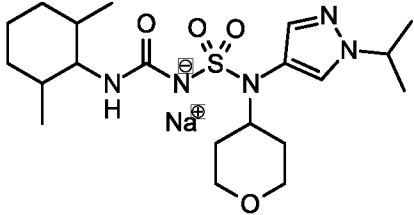
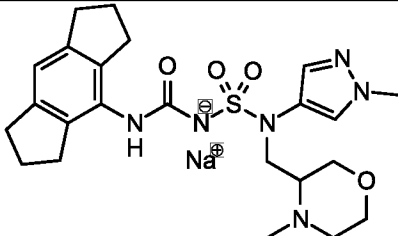
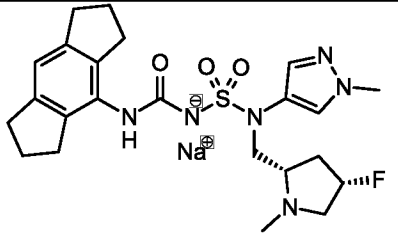
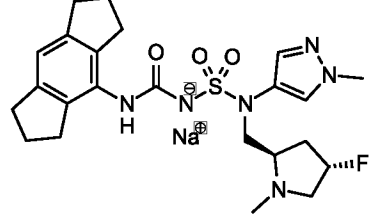
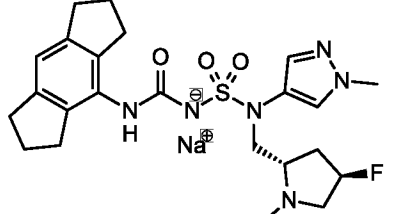
Compound No.	Structure
88	
89	
90	
91	
92	
93	

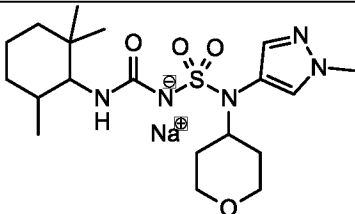
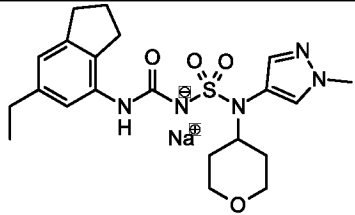
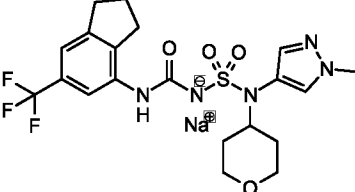
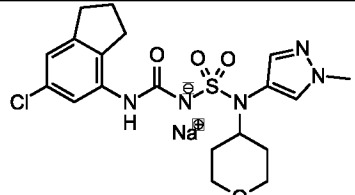
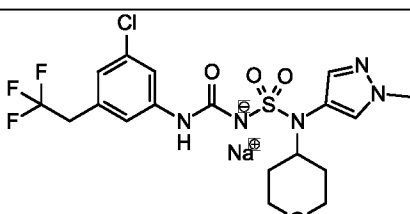
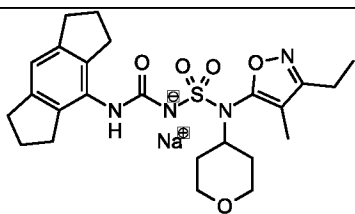
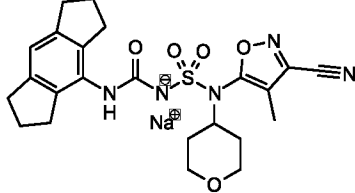
Compound No.	Structure
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95	
96	
97	
98	
99	

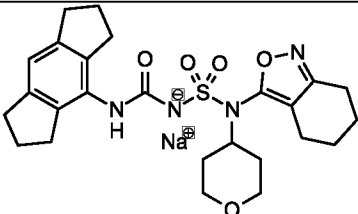
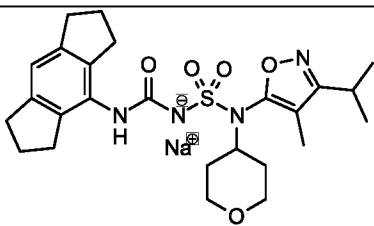
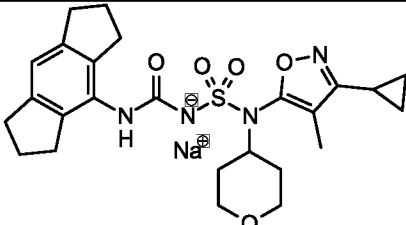
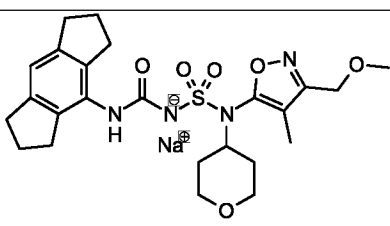
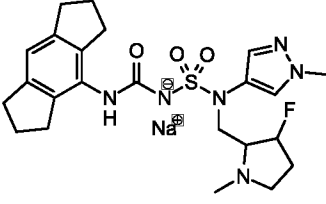
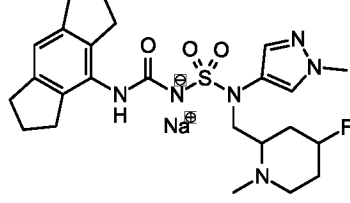
Compound No.	Structure
100	
101	
102	
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104	
105	

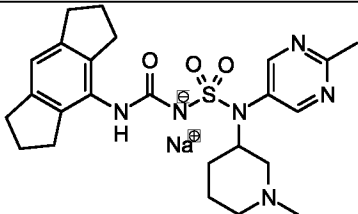
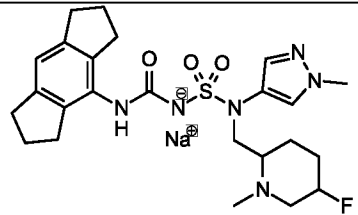
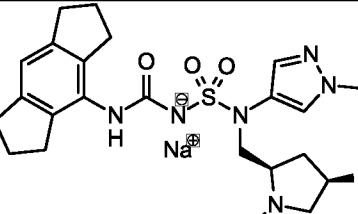
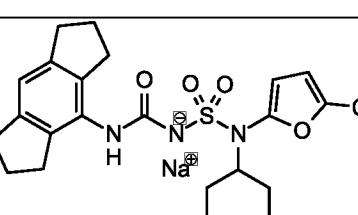
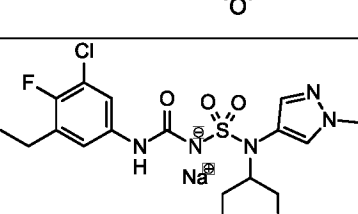
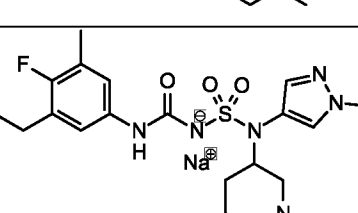
Compound No.	Structure
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107	
108	
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110	
111	

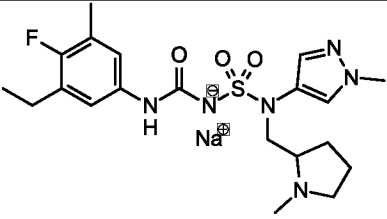
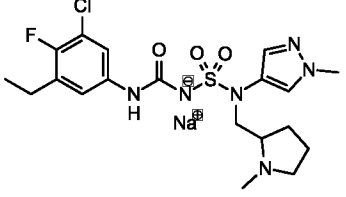
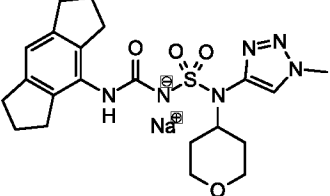
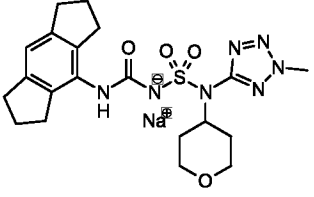
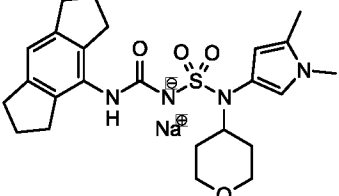
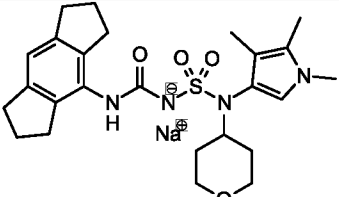
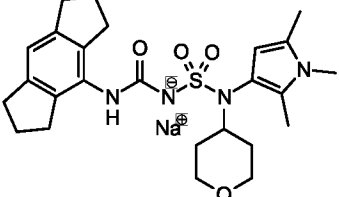
Compound No.	Structure
112	
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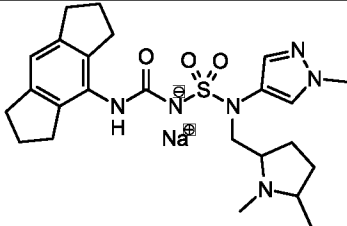
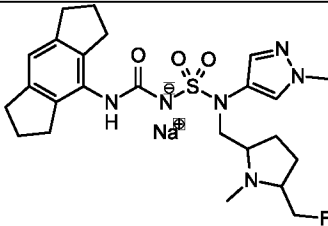
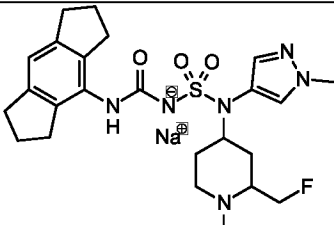
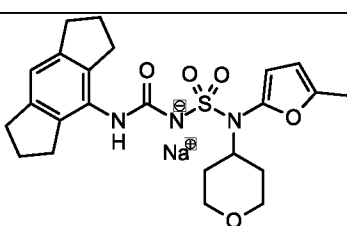
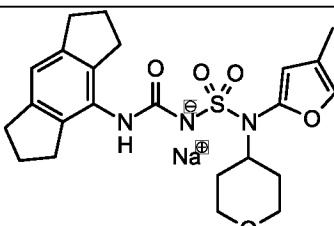
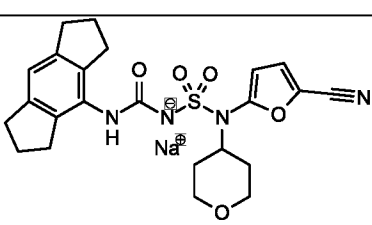
Compound No.	Structure
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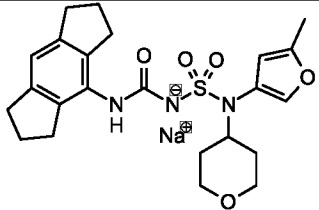
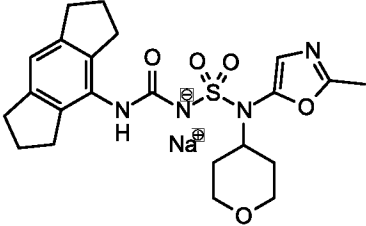
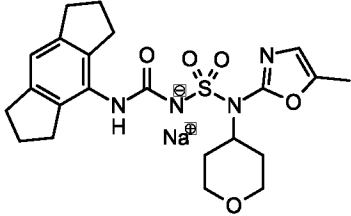
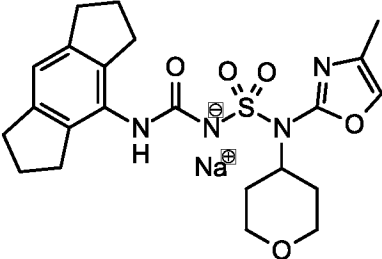
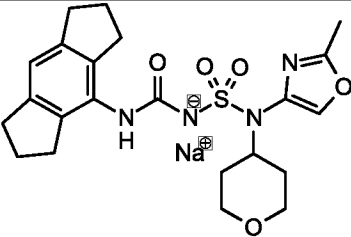
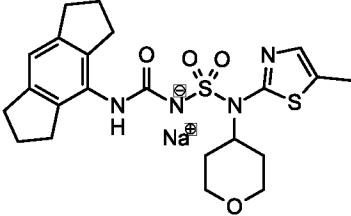
Compound No.	Structure
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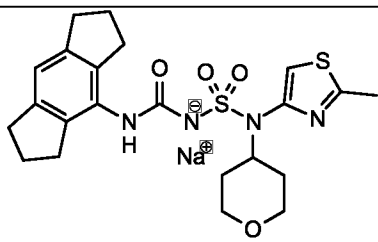
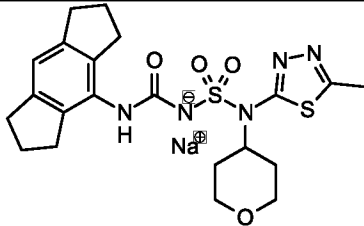
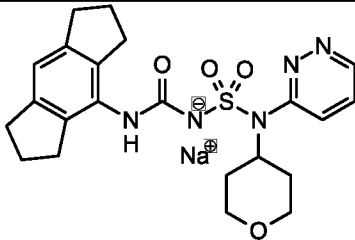
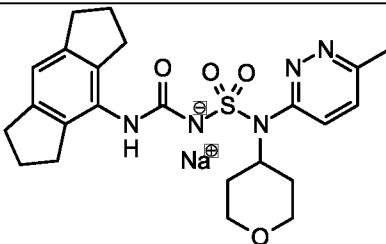
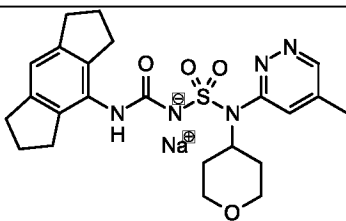
Compound No.	Structure
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Compound No.	Structure
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Compound No.	Structure
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Compound No.	Structure
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Compound No.	Structure
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Compound No.	Structure
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and pharmaceutically acceptable salts thereof.

10. A pharmaceutical composition comprising the compound of any one of claims 1 to 9 or a pharmaceutically acceptable salt thereof, and a pharmaceutically acceptable diluent or carrier.

11. The compound of any one of claims 1 to 9, or the pharmaceutical composition of claim 10,

for use in inhibiting inflammasome activity; optionally, the inflammasome is NLRP3 inflammasome, and the activity is *in vitro* or *in vivo*.

12. The compound of any one of claims 1 to 9, or the pharmaceutical composition of claim 10, for use in treating or preventing a disease or disorder; preferably wherein the disease or disorder is an inflammatory disorder, an autoinflammatory disorder, an autoimmune disorder, a neurodegenerative disease, or cancer.

13. The compound or pharmaceutical composition for use of claim 11 or 12, wherein the disease or disorder is an inflammatory disorder, an autoinflammatory disorder or an autoimmune disorder; optionally, the disease or disorder is selected from cryopyrin-associated autoinflammatory syndrome (CAPS; e.g., familial cold autoinflammatory syndrome (FCAS), Muckle-Wells syndrome (MWS), chronic infantile neurological cutaneous and articular (CINCA) syndrome/ neonatal-onset multisystem inflammatory disease (NOMID)), familial Mediterranean fever (FMF), nonalcoholic fatty liver disease (NAFLD), non-alcoholic steatohepatitis (NASH), gout, rheumatoid arthritis, osteoarthritis, Crohn's disease, chronic obstructive pulmonary disease (COPD), chronic kidney disease (CKD), fibrosis, obesity, type 2 diabetes, multiple sclerosis, dermatological diseases (e.g., acne) and neuroinflammation occurring in protein misfolding diseases (e.g., Prion diseases).

14. The compound or pharmaceutical composition for use of claims 11 or 12, wherein disease or disorder is a neurodegenerative disease; optionally, the disease or disorder is Parkinson's disease or Alzheimer's disease.

15. The compound or pharmaceutical composition for use of claims 11 or 12, wherein the disease or disorder is cancer; optionally, the cancer is metastasising cancer, gastrointestinal cancer, brain cancer, skin cancer, non-small-cell lung carcinoma, head and neck squamous cell carcinoma or colorectal adenocarcinoma.